Array Decomposition

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TBD

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Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All qmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at https://github.com/deleeuw/arrays

1 Introduction

2 Decomposition

2.1 Vectors

Suppose $\mathcal{L}$ is the space of real $n-$element vectors and A is a positive definite matrix of order n. For all $x, y \in \mathcal{L}$ we define the inner product $\langle x, y\rangle := x'Ay$ and for all $x \in \mathcal{L}$ we define the corresponding norm $\|x\|^2 := \sqrt{x'Ax}$.

Suppose $\mathcal{M}$ is subspace of dimension p and X is an orthonormal basis for $\mathcal{M}$, i.e. $n \times p$ matrix with every column in $\mathcal{M}$ and $X'AX = I$. If we want to find the closest point to $y \in \mathcal{L}$ in $\mathcal{M}$ we must minimize $\|y - X\beta\|^2$ over β. The unique minimum is attained at $\widehat{\beta} = X'Ay$ and the projection of y on $\mathcal{M}$ is $X\widehat{\beta} = XX'Ay$.

The projector $P := XX'A$ is a square matrix that is

- idempotent: $P^2 = P$, and
- self-adjoint: for all $x, y \in \mathcal{L}$ $\langle x, Py\rangle = \langle Px, y\rangle$, i.e. $AP = P'A$.

We have $Px = x$ if and only if $x \in \mathcal{M}$. For any projector P the matrix $I - P$ is also a projector, because

- $(I-P)^2 = I - 2P + P^2 = I - P$, and
- $A(I-P) = A - AP = A - P'A = (I-P)'A$.

Moreover P and $(I-P)$ are orthogonal, because $P(I-P) = P - P^2 = 0$.

Suppose $X_\perp$ is an $n \times (n-p)$ orthonormal basis of $\mathcal{M}_\perp$, the subspace of all z such that $z'AX = 0$. Thus

$$\begin{bmatrix} X' \\ X_\perp' \end{bmatrix} A \begin{bmatrix} X & X_\perp \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}, \tag{1a}$$

and consequently

$$\begin{bmatrix} X & X_\perp \end{bmatrix} \begin{bmatrix} X' \\ X_\perp' \end{bmatrix} = A^{-1}. \tag{1b}$$

It follows that $X_\perp X_\perp' A = I - XX'A = I - P$. Thus $I - P$ projects on $\mathcal{M}_\perp$. We have

$$\|Px\|^2 = x'P'APx = x'AP^2x = x'APx = \langle x, Px\rangle, \tag{2}$$

$$\|(I-P)x\|^2 = x'(I-P')A(I-P)x = x'Ax - x'APx = \|x\|^2 - \|Px\|^2, \tag{3}$$

$$\langle Px, (I-P)y\rangle = x'P'A(I-P)y = x'P'Ay - x'P'APy = 0. \tag{4}$$

It is trivially true for any square P that $x = Px + (I - P)x$, but Px and $(I - P)x$ are orthogonal if and only if P is a projector, and in that case we also have Pythagoras

$$\|x\|^2 = \|Px\|^2 + \|(I - P)x\|^2. \tag{5}$$

We can generalize the decomposition $x = Px + (I - P)x$ to several mutually orthogonal subspaces $\mathcal{M}_1, \cdots, \mathcal{M}_m$, with corresponding projectors $P_1, \cdots, P_m$. If $P_1 + \cdots + P_m = I$ we can have the decomposition in orthogonal parts

$$x = P_1x + P_2x + \cdots P_mx, \tag{6}$$

and if $P_1 + \cdots + P_m \lesssim I$ in the Loewner sense[1] then

$$x = P_1x + P_2x + \cdots P_mx + (I - P_1 - \cdots - P_m)x \tag{7}$$

Note that the "residual" $I - P_1 - \cdots - P_m$ is orthogonal to all $P_1, \cdots, P_m$.

Suppose U is an $n \times n$ matrix with $U'U = nI$ and with first column equal to e. Define $\alpha := n^{-1}U'x$ and $P_j := n^{-1}u_ju_j'$. The P_j are projectors orthogonal to each other. Then

$$U\alpha = \sum_{j=1}^{n} P_jx = (\frac{1}{n}ee')x + (I - \frac{1}{n}ee')x = x_\bullet e + (x - x_\bullet). \tag{8}$$

Although α depends on the choice of U the product $U\alpha$ is the same for all U.

We can also derive the decomposition from a least squares projection perspective. Suppose we want to minimize $\|x - a\|^2$ over all a of the form $a = \alpha e$. The solution for α is, of course, $Px = x_\bullet$. And we may want to minimize $\|x - a\|^2$ over all a satisfying $e'a = 0$, which gives the solution $(I - P)x = x - x_\bullet$.

The projection formulation suggests some obvious further generalizations. We can replace the one-dimensional subspace of all constant vectors by an arbitrary subspace S, and we can replace the inner product $x'y$ by an arbitrary inner product $x'Ay$, with A positive semi-definite. If H is an A-orthonormal basis for the subspace S, i.e. if $H'AH = I$, then $Px = HH'Ax$ or $(x - Px)'Az = 0$ for all $z \in S$.

In the next sections we analyse the same approaches we have used for vectors to decompositions of linear spaces of matrices and arrays. This is actually not as much of a generalization as it appears to be, because matrices and arrays are elements of inner product spaces, just like vectors are. What we will introduce are different notations and in most cases different inner products.

[1]The eigenvalues of $P_1 + \cdots + P_m$ are less than or equal to one.

2.2 Matrices

Suppose X is an $n \times m$ matrix, an element of the space $\mathbb{R}^{n\times m}$ with inner product $\langle X, Y\rangle :=$ tr $X'Y$ and squared norm $\|X\|^2 =$ tr $X'X$. The elementwise decomposition we are interested in is

$$x_{ij} = x_{\bullet\bullet} + (x_{i\bullet} - x_{\bullet\bullet}) + (x_{\bullet j} - x_{\bullet\bullet}) + (x_{ij} - x_{i\bullet} - x_{\bullet j} + x_{\bullet\bullet}). \tag{9}$$

Using ANOVA terminology we have the *overall effect*, the two *main effects* for rows and columns, and the *first order interaction* of the rows and columns.

In the language of projectors we define

$$P_r := n^{-1}ee', \tag{10a}$$

$$P_c := m^{-1}ee', \tag{10b}$$

and with $Q_r := I - P_r$ and $Q_c := I - P_c$. Equation (9) becomes

$$X = P_rXP_c + P_rXQ_c + Q_rXP_c + Q_rXQ_c. \tag{11}$$

Thus again we have four projectors, orthogonal to each other, adding up to the identity.

In de Leeuw (2026) we generalize the results in this section to the inner product $\langle X, Y\rangle :=$ tr $RXCY'$, with R and C positive semi-definite. Decomposition (11) remains correct, with $P_r = ee'R/e'Re$ and $P_c = Cee'/e'Ce$. Note that both projectors are asymmetric. Equation (9) also remains valid, if we interpret the bullitt subscripts as weighted means.

To see more clearly how the vector case and the matrix case are different we use the vec() operator to map $\mathbb{R}^{n\times m}$ into $\mathbb{R}^{nm}$. Using the Kronecker product $\otimes$ of matrices we have the identity

$$\text{tr } RXCY' = x'(C \otimes R)y, \tag{12}$$

with $x := \text{vec}(X)$ and $y = \text{vec}(Y)$. Thus our matrix results can be interpreted as a special case of the vector results in $\mathbb{R}^{nm}$, with inner product $x'(C \otimes R)y$.

Now we switch to using orthogonal functions on a product space. This part is inspired by an anonymous and unpublished manuscript that I read around 1980 with a title that might have been "Lois de Probabilité sur un Ensemble Produit" and perhaps the words "Décomposition" and/or "Interaction" were also in the title or subtitle. It was either by Jean-Paul Benzécri or by one of his students or co-workers. I did have a copy and I lost it, and Google can't find me a new one. The published paper by Bener (1982) discusses the same materials, but in less detail.

Define U and V to be square matrices with first column equal to e, $U'U = nI$ and $V'V = mI$. Note this implies $UU' = nI$ and $VV' = mI$, and thus $UU' - ee' = nJ_n$ and $VV' - ee' = mJ_m$. Define $\alpha_{ij} := (nm)^{-1}u_i'Xv_j$. Then

$$a_{ij}u_iv_j' = P_i^UXP_j^V, \tag{13}$$

where $P_i^U := n^{-1}u_iu_i'$ and $P_j^V := \frac{1}{m}v_jv_j'$ project on the lines through the origin and respectively u and v. Using these projectors we have the identity

$$X = \sum_{i=1}^{n}\sum_{j=1}^{m}\alpha_{ij}u_iv_j' = \sum_{i=1}^{n}\sum_{j=1}^{m}P_i^UXP_j^V, \tag{14}$$

and $\|X\|^2 = \|A\|^2$, where $A := \{\alpha_{ij}\}$. If $i \neq k$ then $P_i^UP_k^U = 0$ and if $j \neq l$ then $P_j^VP_l^V = 0$. Moreover if I and K are disjoint subsets of $\{1, \cdots, n\}$ then

$$\left(\sum_{i\in I}P_i^U\right)\left(\sum_{k\in K}P_k^U\right) = 0. \tag{15}$$

Although P_i^U and P_j^V are projectors on $\mathbb{R}^n$ and $\mathbb{R}^m$, they are only the building blocks for the projectors on $\mathbb{R}^{n\times m}$ that we use in this section. Define

$$\Pi_{ij}(X) := P_i^UXP_j^V \tag{16}$$

then Π_{ij} projects $\mathbb{R}^{n\times m}$ on the line through the origin and u_iv_j'. Thus $\Pi_{ij}\circ\Pi_{ij} = \Pi_{ij}$ and $\Pi_{ij}\circ\Pi_{kl} = 0$ unless $i = k$ and $j = l$. Also

$$\|X\|^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}\|\Pi_{ij}(X)\|^2 \tag{17}$$

$$\Pi_{11}(X) = x_{\bullet\bullet}ee'. \tag{18a}$$

$$\sum_{j\neq 1}^{m}\Pi_{1j}(X) = e(x_{\bullet_j} - x_{\bullet\bullet})', \tag{18b}$$

$$\sum_{i\neq 1}^{m}\Pi_{i1}(X) = (x_{i\bullet} - x_{\bullet\bullet})e', \tag{18c}$$

$$\sum_{i\neq 1}^{n}\sum_{j\neq 1}^{m}\Pi_{ij}(X) = J_nXJ_m = (x_{ij} - x_{i\bullet}e' - ex_{j\bullet}' + x_{\bullet\bullet}) \tag{18d}$$

Although the Π_{ij} depend on the choice of U and V the partial sums defining the main effects and the interaction are independent of U and V.

2.3 Arrays

If X is an $n \times m \times \ell$ array then the basic elementwise decomposition we are interested in is

$$\begin{aligned} x_{ijk} = x_{\bullet\bullet\bullet} &+ (x_{i\bullet\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet j\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) && (19) \\ &+ (x_{ij\bullet} - x_{i\bullet\bullet} - x_{\bullet j\bullet} + x_{\bullet\bullet\bullet}) && (20) \\ &+ (x_{i\bullet k} - x_{i\bullet\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) && (21) \\ &+ (x_{\bullet jk} - x_{\bullet j\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) && (22) \\ &+ (x_{ijk} - x_{ij\bullet} - x_{i\bullet k} - x_{\bullet jk} + x_{i\bullet\bullet} + x_{\bullet j\bullet} + x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) && (23) \end{aligned}$$

3 Correspondence Analysis

Suppose F is a two-dimensional $n \times m$ contingency table, normalized to have grand total equal to one, with row marginals in a diagonal matrix D and column marginals in a diagonal matrix E. Use the inner product $\langle F, G\rangle = \text{tr } D^{-1}FE^{-1}G'$. The projectors are

$$f_{ij}/d_ie_j = 1 + \sum_{s=1}^{m} u_{is}v_{js}$$

4 Log-linear Analysis

References

Bener, A. 1982. "Décomposition des Interactions dans une Correspondance Multiple." *Les Cahiers de l'Analyse Des Données* 7 (1): 25–32. https://www.numdam.org/item/CAD_1982__7_1_25_0.pdf.

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